CSC5041 Assignment1

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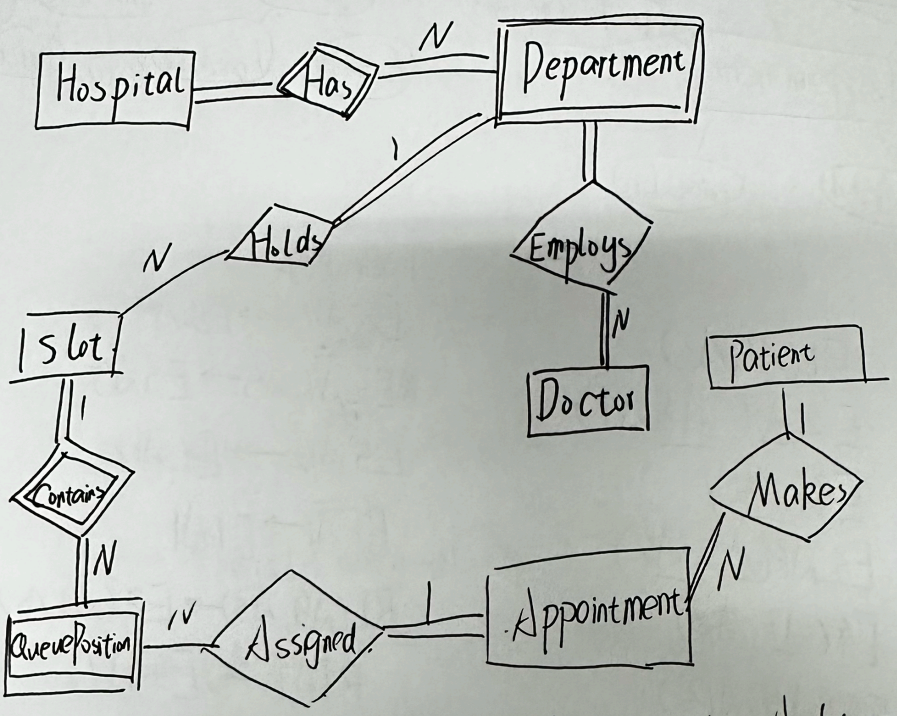
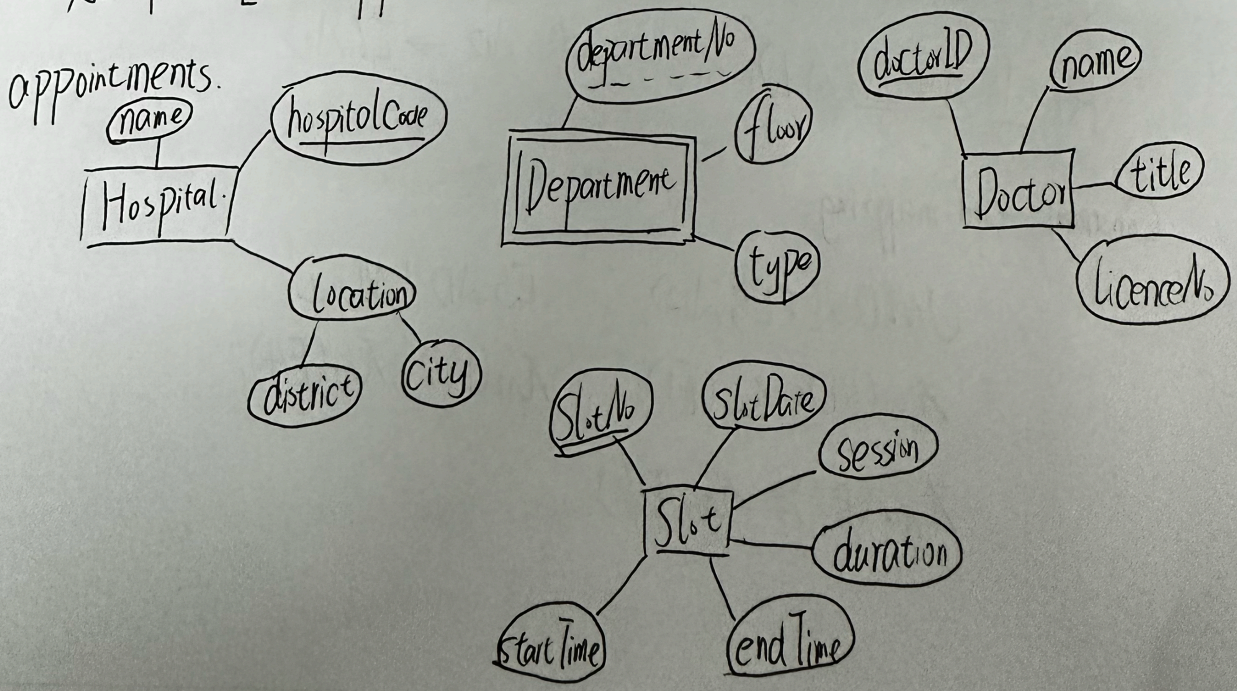
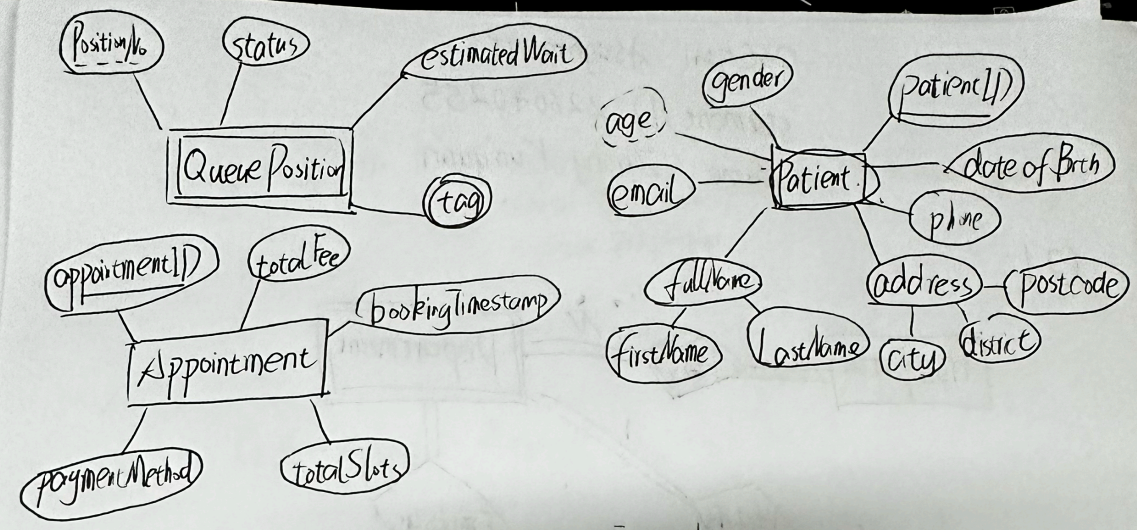
Q1:



Assumptions: Each appointment contains at least one slot. A slot may have no appointments.





Q2

E1(A1, A2)

E2(A9, A3, A4)

E3(A5)

E3_A6(A5, A6)

E4(A7, A8)

E5(A9, A12)

E6(A12, A14, A15)

R1(A1, A9, A3, A7)

Forein keys:

E2.A9 → E5.A9

E3_A6.A5 → E3.A5

E5.A12 → E6.A12

R1.A1 → E1.A1

R1.(A9, A3) → E2.(A9, A3)

R1.A7 → E4.A7

R3.A5 → E3.A5

R3.A12 → E6.A12

Constrains and mapping:

UNIQUE(E5.A12), E5.A12 ≠ NULL

$\tau_{A1}(R1) = \tau_{A1}(E1)$, $\tau_{A7}(R1) = \tau_{A7}(E4)$;

$\tau_{A12}(R3) = \tau_{A12}(E6)$.

Q3

(1) $\pi_{RiderName}\left(Rider \bowtie \sigma_{category='car' \wedge tripDate>'2025-01-01' \wedge tripDate<'2025-07-01'}(Trip \bowtie VehicleType)\right)$

(2)

$G \leftarrow \pi_{riderID} - \pi_{riderID}(\sigma_{fare>10}(Trip))$,

$Answer: \leftarrow \gamma_{riderID;\ MAX(amount)}(G \bowtie Payment)$

(3)

$C \leftarrow \pi_{category}(VehicleType)$

$U \leftarrow \pi_{riderID,\ category}(Trip \bowtie VehicleType)$

$G \leftarrow U \div C$

$Answer: \pi_{riderID,\ memberTier}(Rider \bowtie G)$

(4)

$S \leftarrow \rho_{(startStationID,\ zoneID,\ capacity,\ stationType)}(Station)$,

$T \leftarrow \sigma_{zoneName='Xuhui' \wedge tripDate \geq '2025-10-01' \wedge tripDate \leq '2025-11-02'}(Trip \bowtie S \bowtie Zone \bowtie Rider)$,

$V \leftarrow \pi_{tripID,\ age}(T)$

$Answer \leftarrow \gamma_{AVG(age)}(V)$